

Cover Letter

September 16, 2026

To the Editor-in-Chief, Prof. Minvydas Ragulskis
Mathematical Models in Engineering
Extrica (JVE International Ltd.)

Dear Editor,

Please consider our manuscript, "**Edge-Deployable EEG Foundation Models: Scaling Laws and Cross-Dataset Transfer for Compact SIGReg-Pretrained Encoders**," for publication as an original research article in *Mathematical Models in Engineering*.

Self-supervised EEG foundation models have advanced rapidly, but the leading systems use 5–10 M+ parameters and multi-GPU training budgets aimed at cross-dataset generalization. The complementary regime that most laboratories and on-device applications actually operate in — focused, single-corpus pretraining for in-distribution tasks under a tight compute budget — remains largely uncharacterized. Our work addresses that gap directly. We present EEG-LeJEPA, a 2.86 M-parameter Joint-Embedding Predictive Architecture trained with Sketched Isotropic Gaussian Regularization, whose entire pretraining-plus-evaluation pipeline runs on a single consumer GPU in under 15 minutes at a compute cost below one US dollar.

We believe the manuscript is well matched to *Mathematical Models in Engineering*, whose principal topics include artificial intelligence and machine learning, detection of objects and patterns, and formal mathematical models applied to engineering problems. Our work is built on an explicit mathematical model: the Sketched Isotropic Gaussian Regularizer casts representation learning as matching the embedding distribution to an isotropic Gaussian, a constraint made tractable through the Cramér–Wold theorem and scored with the closed-form Cramér–von Mises statistic on random one-dimensional projections. We characterize this model's behaviour empirically — its scaling law across data and compute, its sensitivity to the projection count relative to embedding dimension, and its representational collapse when the regularizer is removed — and apply it to a practical engineering problem: pattern detection in EEG for brain–computer interfaces on resource-constrained hardware. Beyond the headline results, the paper contributes findings of methodological interest to the journal's readership:

- A scaling-law characterization across six (data, compute) operating points: best-source AUC rises then plateaus near 0.76, and a 2.4× larger model is modestly but consistently better at matched data (seed-stable across three seeds) — the plateau reflects a capacity limit, not a data limit. An isolation run further shows the SIGReg projection count must scale with embedding dimension; ignoring this manufactures a spurious "larger model regresses" result, a previously unreported practical pitfall when applying isotropy regularizers across model widths.
- A SIGReg-weight ablation revealing a sharp optimum and **complete representational collapse** when the regularizer is removed, giving a clean empirical validation of the anti-collapse mechanism.
- A counterintuitive cross-dataset transfer result: channel-padded transfer from a full-montage encoder **outperforms** channel-matched re-pretraining.

- A leakage-clean evaluation protocol: we show that allowing evaluation subjects into pretraining inflates predictor-derived feature sources, and under a strictly held-out split the simple encoder mean-pool is both the strongest and the most trustworthy source.
- A controlled comparison showing the self-supervised, frozen-encoder probe **ties** end-to-end supervised training on left-versus-right MI (-0.2 pp, n.s.) and **significantly exceeds** it on rest-versus-activity ($+8.4$ pp, $p < 0.001$), with one frozen encoder serving both tasks.

On our two downstream EEGMMIDB tasks the model reaches 68.6% accuracy / 0.751 AUC on rest-versus-activity and 62.1% / 0.690 AUC on left-versus-right motor imagery (encoder mean-pool features) via linear probe, under strict leave-one-subject-out evaluation with the 20 evaluation subjects held out of pretraining (both gains over a random-initialized baseline significant, $q < 0.002$). All headline differences are assessed with paired significance tests (exact sign-flip permutation tests, bootstrap confidence intervals, and Benjamini–Hochberg correction), and the complete pipeline — code, configurations, trained checkpoints, and per-fold outputs — is released for full reproducibility at <https://github.com/lifeinpassion/eeg-lejepa>.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All data are from public repositories (PhysioNet EEGMMIDB and BCI Competition IV Dataset 2a); no new human-subjects data were collected. The authors declare no competing interests.

We note for transparency that an independent and contemporaneous study (Laya; Panchavati et al.) applies the same family of methods to EEG at foundation scale; our contribution is complementary, addressing the compact, single-corpus, edge-deployment regime and the scaling, ablation, and transfer questions that work does not examine.

Thank you for your consideration. We look forward to your response.

Sincerely,

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